

TURKMENABAT, Turkmenistan

WMO: 386870

Lat: 39.083N

Lon: 63.600E

Elev: 190

StdP: 99.06

Time Zone: 5.00 (E05)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-10.6	-7.2	-15.4	1.0	-7.7	-12.2	1.3	-3.7	9.5	12.5	8.4	9.2	2.5	0	0.486

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.2	39.7	20.6	38.2	20.2	36.9	19.9	22.6	35.5	21.7	34.6	20.9	33.9	4.1	20

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.4	13.6	28.6	17.2	12.6	27.7	16.1	11.7	27.3	67.9	35.4	64.3	34.7	61.4	33.9	28.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
8.4	7.3	6.4	DB	-12.6	42.1	4.1	1.4	-15.6	43.1	-18.0	43.9	-20.4	44.7	-23.4	45.7	
			WB	-13.1	25.0	4.1	1.4	-16.1	26.0	-18.5	26.9	-20.8	27.7	-23.8	28.7	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	16.4	3.1	5.2	11.5	17.9	23.7	27.9	29.2	27.4	21.9	15.5	8.7	3.9
	DBStd	10.27	5.26	5.66	5.06	4.76	3.31	2.62	2.55	2.71	3.37	4.39	5.11	4.86
	HDD10.0	699	219	150	43	3	0	0	0	0	0	8	80	196
	HDD18.3	1978	472	369	217	64	3	0	0	0	9	108	289	447
	CDD10.0	3035	5	14	90	240	426	536	597	541	357	179	42	8
	CDD18.3	1272	0	0	6	51	171	286	338	283	116	20	2	0
	CDH23.3	17061	0	1	80	581	2066	4037	4863	3743	1377	288	26	0
	CDH26.7	9559	0	0	17	214	1003	2400	3013	2210	622	76	4	0
Wind	WSAvg	3.1	2.9	3.3	3.4	3.3	3.1	3.3	3.5	3.3	2.9	2.5	2.5	2.8
Precipitation	PrecAvg	127	18	20	27	21	9	1	0	0	1	3	12	16
	PrecMax	218	64	59	64	92	57	9	3	3	23	19	44	67
	PrecMin	47	0	2	2	0	0	0	0	0	0	0	0	0
	PrecStd	39	15	15	16	19	12	2	1	1	3	5	10	13
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	19.3	21.8	29.0	35.0	37.8	40.5	42.4	40.4	36.3	32.0	26.5	19.9
		MCWB	9.7	11.7	15.3	17.6	19.7	20.9	20.9	20.6	19.3	17.5	14.5	9.9
	2%	DB	15.8	18.7	25.7	31.6	35.7	38.7	40.2	38.7	34.3	28.7	22.6	16.3
		MCWB	8.5	10.3	14.2	17.0	18.7	20.2	21.1	20.3	18.5	15.8	13.0	8.9
	5%	DB	13.1	16.0	22.8	29.1	34.0	37.4	38.6	37.1	32.6	26.6	19.5	13.9
		MCWB	7.7	9.3	13.0	16.0	18.1	19.7	20.5	20.2	17.8	15.0	11.6	8.1
	10%	DB	10.4	13.5	20.1	26.8	32.1	36.0	37.0	35.7	30.4	24.3	16.9	11.2
		MCWB	6.4	8.3	12.1	15.2	17.6	19.3	20.2	19.4	16.9	13.8	10.4	6.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.7	12.8	16.5	19.8	22.1	23.1	24.6	23.2	21.2	18.6	15.5	11.2
		MCDB	16.3	18.9	25.4	29.4	33.2	36.6	37.3	35.5	33.1	29.1	23.6	16.7
	2%	WB	9.3	11.3	14.9	18.3	20.3	21.9	22.9	22.1	19.7	16.7	13.8	9.8
		MCDB	14.5	17.1	23.0	28.5	31.9	34.8	35.8	34.9	31.6	26.7	21.2	15.1
	5%	WB	8.0	10.0	13.7	17.1	19.3	20.9	21.9	21.1	18.5	15.6	12.3	8.4
		MCDB	12.5	14.9	21.3	26.4	30.8	34.2	34.9	33.9	30.3	24.7	18.3	13.0
	10%	WB	6.6	8.6	12.5	16.1	18.4	20.1	21.1	20.3	17.5	14.6	11.0	7.2
		MCDB	10.1	13.0	19.3	24.8	29.4	33.3	34.3	33.3	29.0	23.0	16.1	10.9
Mean Daily Temperature Range	5% DB	MDBR	9.0	10.2	11.4	12.7	13.9	14.6	14.2	15.2	15.6	14.6	11.6	9.1
		MCDBR	13.2	13.9	15.5	16.2	16.5	16.6	15.9	16.9	17.8	17.1	15.1	12.9
		MCWBR	7.3	7.2	6.5	5.6	5.5	5.7	5.5	6.2	7.0	7.4	7.1	7.2
	5% WB	MCDBR	12.1	12.3	14.0	14.1	14.5	15.4	14.9	16.0	17.0	16.0	13.7	11.8
		MCWBR	7.2	6.8	6.3	5.5	5.7	5.8	5.7	6.5	7.1	7.3	6.9	7.1
Clear-Sky Solar Irradiance	taub	0.383	0.427	0.500	0.531	0.529	0.531	0.500	0.458	0.437	0.474	0.444	0.390	
	taud	2.112	2.001	1.851	1.816	1.847	1.844	1.922	2.018	2.032	1.912	1.969	2.079	
	Ebn at Noon	749	766	748	756	768	766	786	810	799	704	665	702	
	Edh at Noon	114	146	188	207	204	205	188	167	154	157	131	110	
All-Sky Solar Radiation	RadAvg	2.10	3.07	4.23	5.63	6.92	7.85	7.77	7.13	5.84	4.06	2.59	1.88	
	RadStd	0.19	0.31	0.32	0.44	0.28	0.23	0.16	0.15	0.15	0.23	0.24	0.20	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.45	N/A	N/A	+0.78	-0.54	-0.64	N/A	N/A	+130	+128
(47)	Regional (8 neighbors)	N/A	N/A	N/A	+0.31	-0.68	-0.92	N/A	N/A	N/A	+50

Nomenclature: See separate page